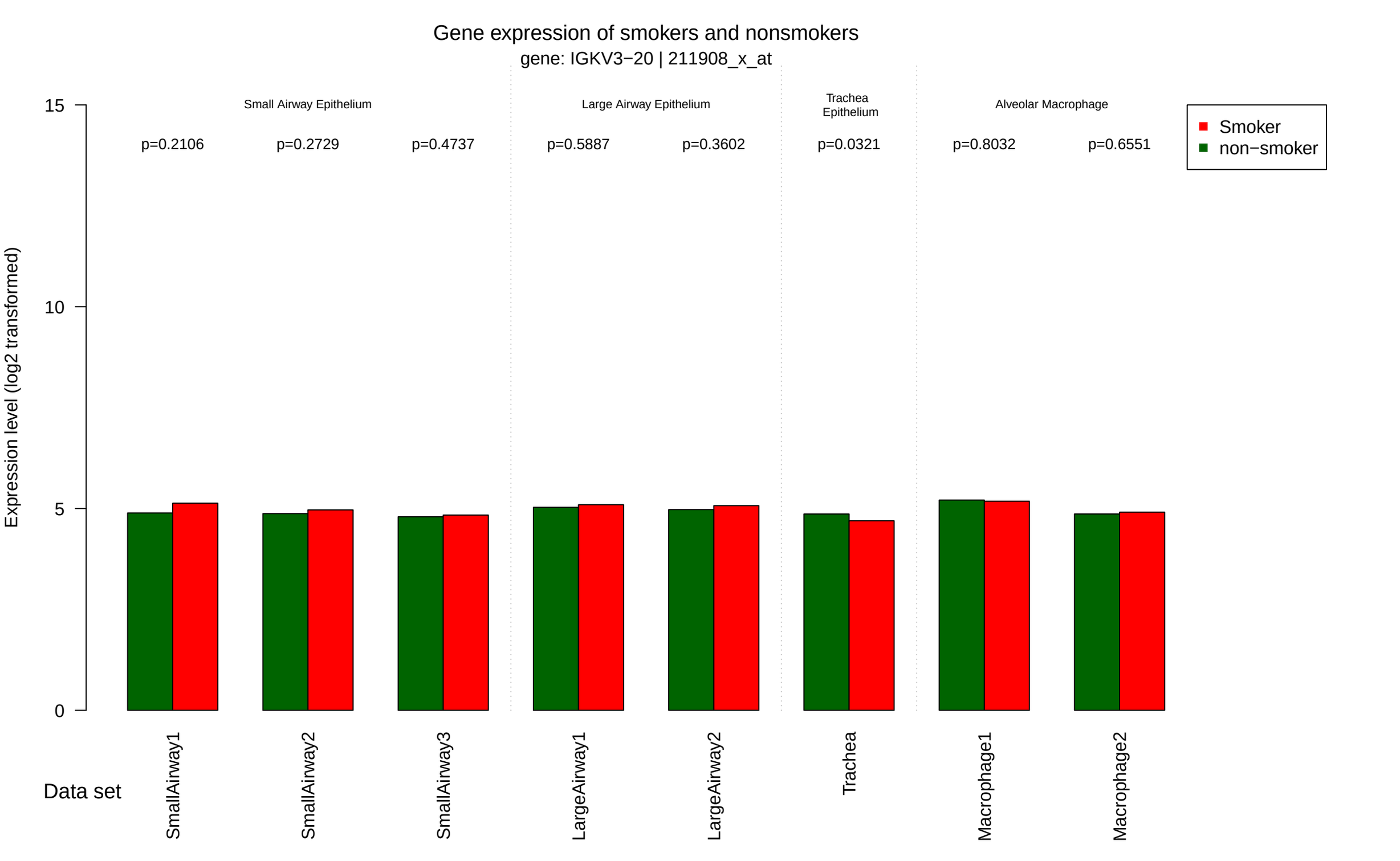
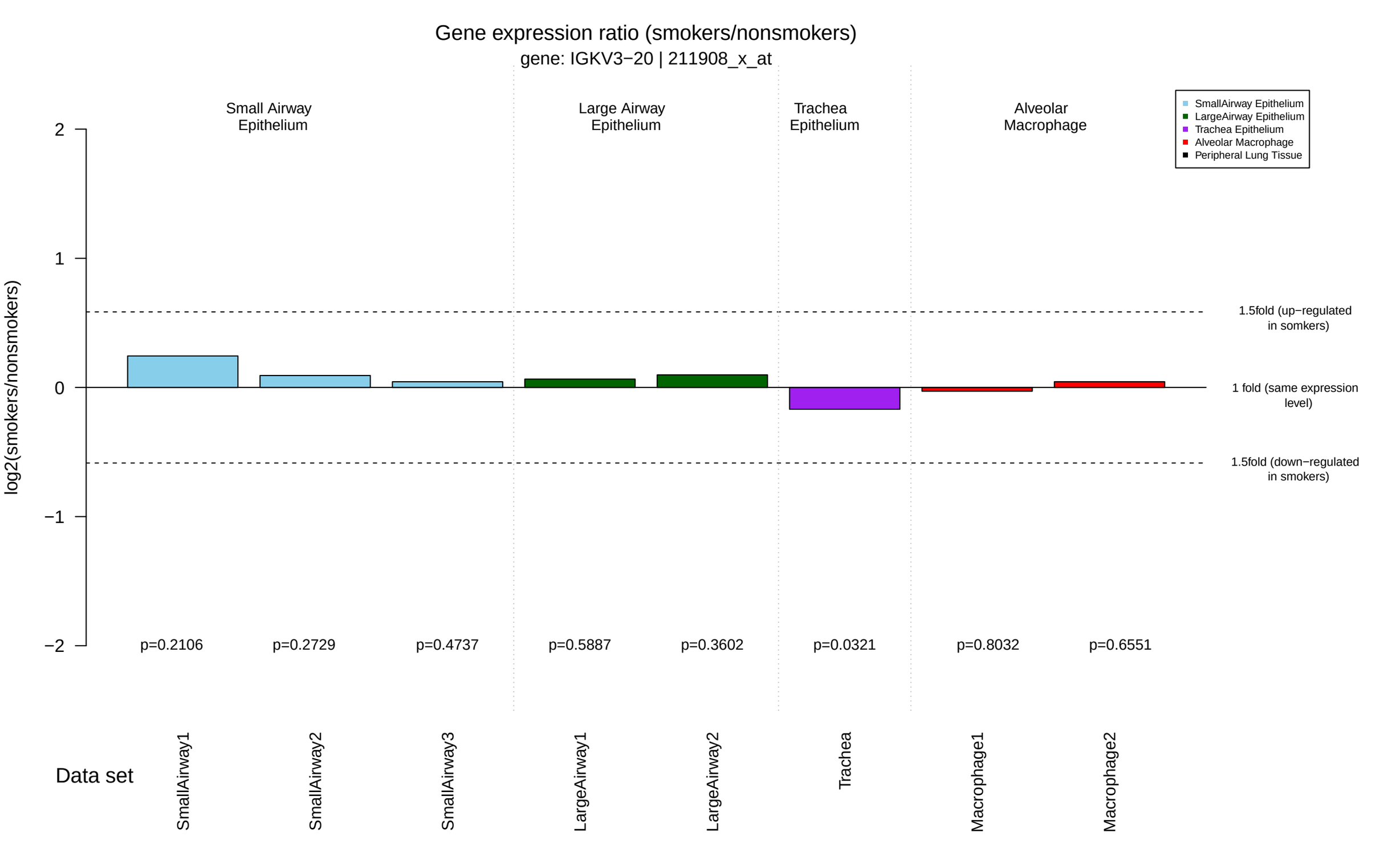
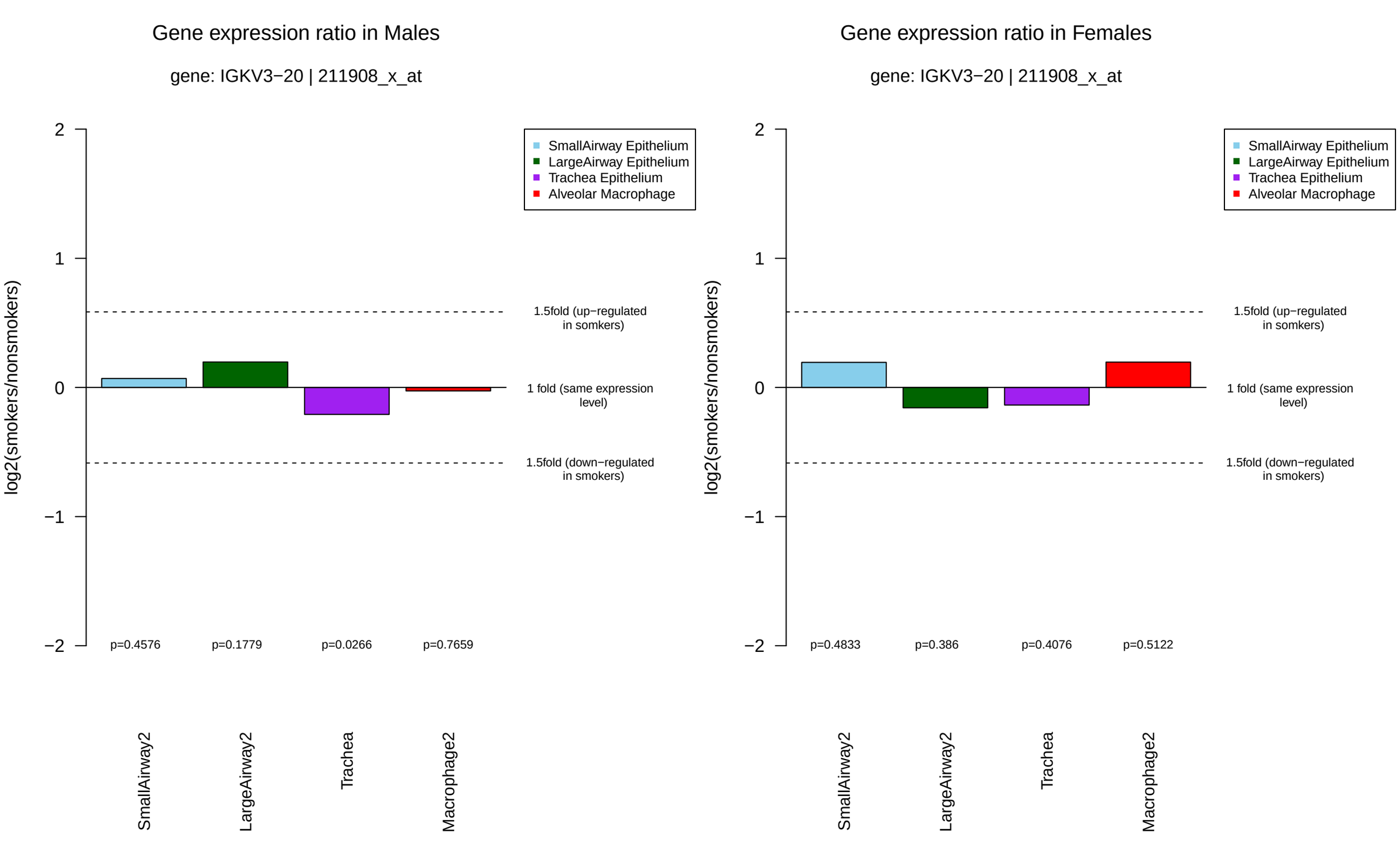
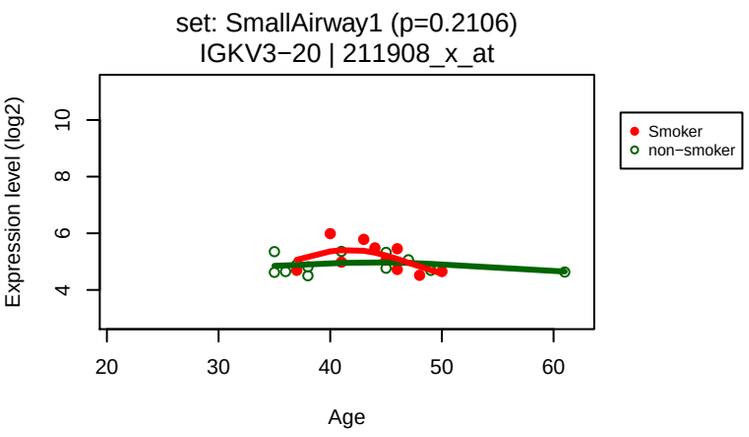
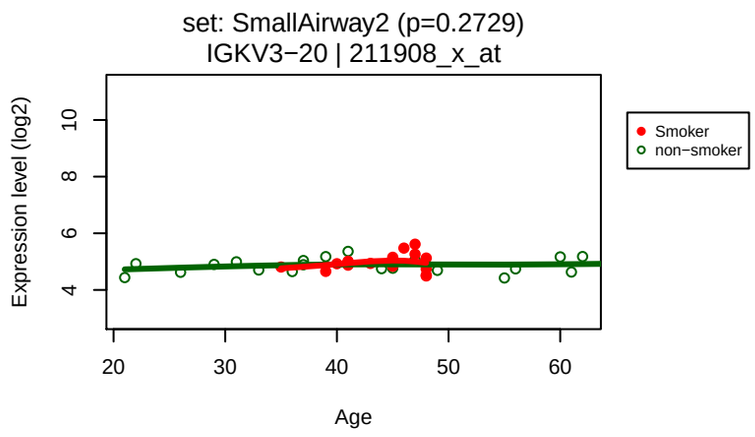
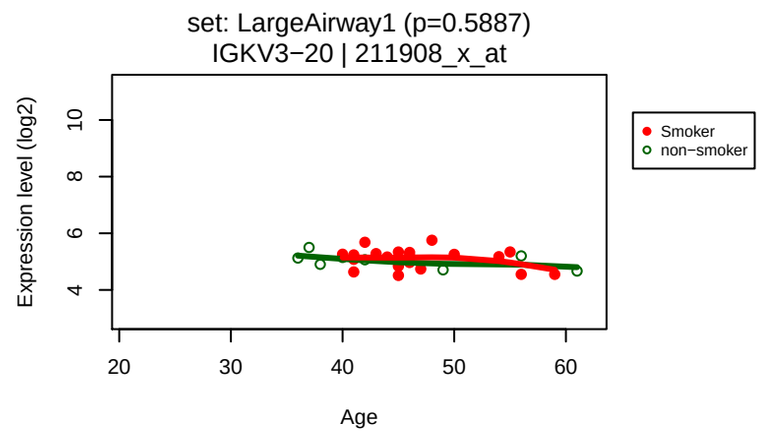

Gene expression of smokers and nonsmokers
gene: IGKV3−20 | 211908_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2106
p=0.2729
p=0.4737
p=0.5887
p=0.3602
p=0.0321
p=0.8032
p=0.6551
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

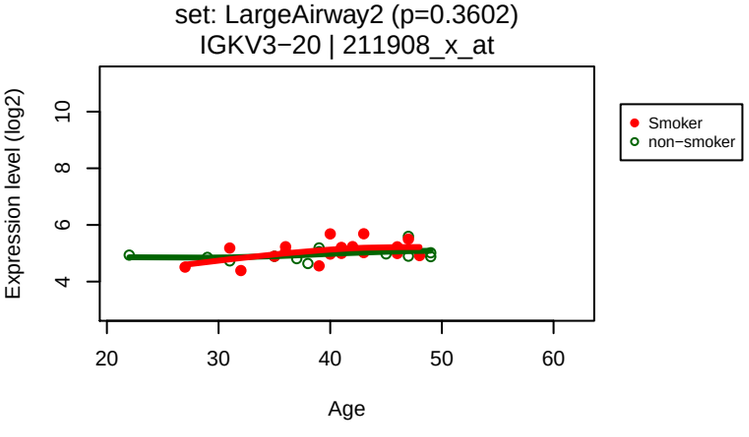

Gene expression ratio (smokers/nonsmokers)
gene: IGKV3−20 | 211908_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2106
p=0.2729
p=0.4737
p=0.5887
p=0.3602
p=0.0321
p=0.8032
p=0.6551
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

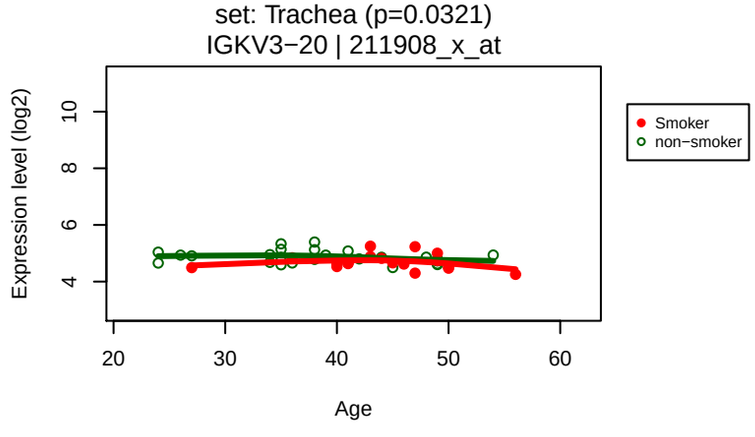

Gene expression ratio in Males
gene: IGKV3−20 | 211908_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4576
p=0.1779
p=0.0266
p=0.7659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGKV3−20 | 211908_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4833
p=0.386
p=0.4076
p=0.5122
SmallAirway2
LargeAirway2
Trachea
Macrophage2

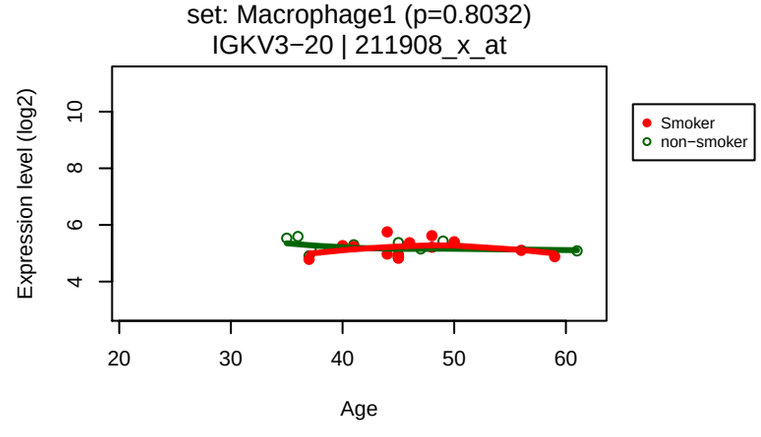

set: SmallAirway1 (p=0.2106)
IGKV3−20 | 211908_x_at
Expression level (log2)
Age
Smoker
non−smoker

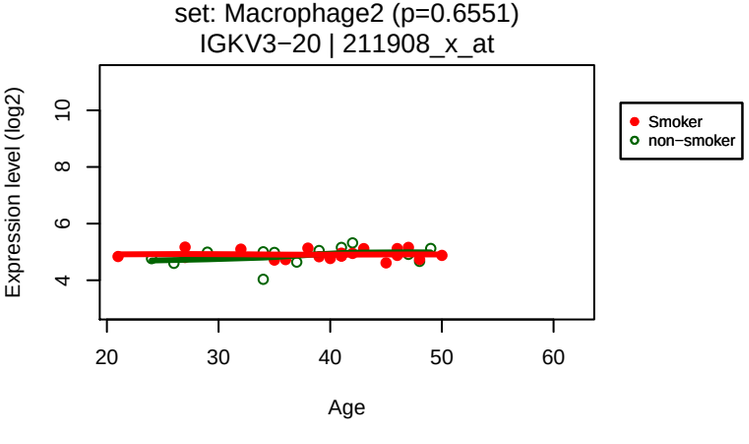

set: SmallAirway2 (p=0.2729)
IGKV3−20 | 211908_x_at
Expression level (log2)
Age
Smoker
non−smoker

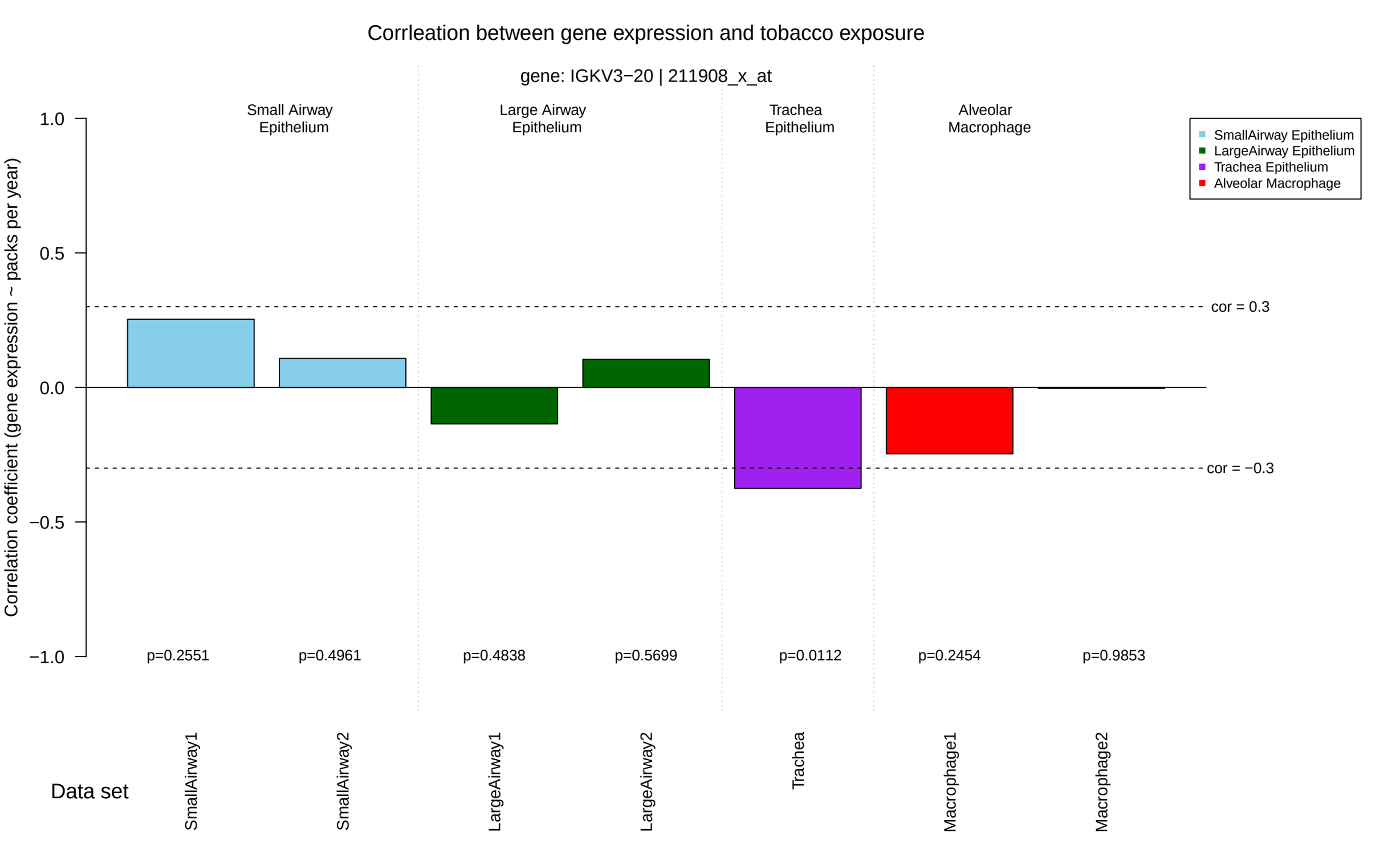

set: LargeAirway1 (p=0.5887)
IGKV3−20 | 211908_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3602)
IGKV3−20 | 211908_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0321)
IGKV3−20 | 211908_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8032)
IGKV3−20 | 211908_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6551)
IGKV3−20 | 211908_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGKV3−20 | 211908_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2551
p=0.4961
p=0.4838
p=0.5699
p=0.0112
p=0.2454
p=0.9853
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2